

Distributed Systems - Notes - v0.1.0

260236

September 2026

Preface

Every theory section in these notes has been taken from the sources:

- Course slides. [1]

About:

 [GitHub repository](#)



These notes are an unofficial resource and shouldn't replace the course material or any other book on distributed systems. It is not made for commercial purposes. I've made the following notes to help me improve my knowledge and maybe it can be helpful for everyone.

As I have highlighted, a student should choose the teacher's material or a book on the topic. These notes can only be a helpful material.

Contents

1	Introduction	4
1.1	Why Distributed Systems?	4
1.1.1	Motivation: Why Distributed Systems?	4
1.1.2	Why Distribute a System?	14
	Index	19

1 Introduction

1.1 Why Distributed Systems?

1.1.1 Motivation: Why Distributed Systems?

There is a very simple reason why distributed systems are important: most applications we use today are **distributed**. Many applications cannot realistically run as a single program on a single computer.

Example 1: Distributed Applications

E-mail, web/cloud applications such as *Google Maps*, *Google Docs*, *Microsoft Office 365*; mobile applications and mobile devices, web services and websites, and many more, are all distributed systems.

E-mail is used as an especially clear example because sending a message involves several independent machines communicating over a network:

User A → mail server A → network → mail server B → User B

There is no single computer performing the entire operation.

So the first intuition is that **modern applications are naturally composed of multiple communicating machines**. This is why distributed systems are not some niche branch of computer science, they are behind a huge fraction of modern computing infrastructure.

State of the Internet

Censys on 2023 created a report about the **observable public Internet**, built from large-scale Internet scanning.¹ The report says that in one daily snapshot in early 2023, Censys observed more than **740 million hosts** exposing about **1.3 billion HTTP services**, roughly **18%** hosted by major cloud providers such as Amazon, Oracle, Google Cloud, or Azure. And more than **85%** of HTTP services running on named hosts.

 A distributed system may therefore consist not of two or three machines, but potentially of 10^3 , 10^5 , 10^6 , or even more interacting components. But this immediately hints at problems that will appear later in the notes: scalability, failures, communication, naming, synchronization, replication, and so on. But at this point, we are just trying to understand why distributed systems are important.

Large-scale Web traffic

From the same Censys report, we can also see the **traffic** perspective rather than the number of machines. For example, the report says that in 2023, on Google:

¹Censys is a platform that **scans and indexes Internet-connected systems and services**. It is like a search engine for Internet infrastructure.

- More than **100k searches per second**
- More than **100 billion visits per month**
- YouTube as another extremely large service
- Google and YouTube producing more traffic than many other major websites combined

Again, the exact values are secondary. The important question is: *could a single server handle all of this?* Clearly not. A large Web service therefore has to distribute work across many machines. And this is the first glimpse of the **scalability** problem that distributed systems have to solve.

In summary, **modern computing operates at a scale where distribution is not just a design choice, but unavoidable.** We saw how widespread distributed applications are, the size of the Internet, and finally the traffic generated by huge Web services.

🌲 LLM infrastructures as distributed systems

Let's consider what happens when we send a query to a modern Large Language Model (LLM) such as ChatGPT. It is tempting to imagine:

Our request → one very powerful GPU → our answer

But this is not how it works. Modern LLMs are too large to fit on a single GPU. So a single query to a modern LLM may involve roughly **16 to 32 GPUs**, corresponding to a couple of multi-GPU nodes. Each GPU may consume up to about **700 W**.

So even a single logical request may require computation spread across several physical processors:

$$\text{Request} \rightarrow \begin{cases} \text{GPU}_1 \\ \text{GPU}_2 \\ \text{GPU}_3 \\ \vdots \\ \text{GPU}_{16-32} \end{cases} \rightarrow \text{Answer}$$

From the user's perspective, however, we do not see any of this. We simply **interact with one service: LLM service.** This is already one of the central ideas of distributed systems: **many independent computational resources cooperate to provide what appears to the user as one single service.** The infrastructure becomes even more clearly distributed when we move from one request to the whole data center. An AI data center may contain **more than 100,000 AI accelerators**, with power consumption potentially reaching hundreds of megawatts.

🧩 So there are actually **two different forms of distribution** here:

- At the level of a **single request**:

One Computation $\rightarrow$ Multiple GPUs

- At the level of the **service as a whole**:

Millions of requests $\rightarrow$ Many machines $\rightarrow$ Many clusters
 $\rightarrow$ Possibly many data centers

The reason this example matters is that distribution is not only about connecting independent applications. Sometimes even **one computation is too large to execute efficiently on one processor**. That naturally connects distributed systems with high-performance computing (HPC), although the two are not identical. However, **modern AI services are distributed both because of computational size and service scale**.

Google as a distributed “supercomputer”

Eric Schmidt (CEO of Google from 2001 to 2011) described Google as essentially “*a large distributed supercomputer spread across many data centers*”. That is a very useful mental model. Instead of imaging Google as a single website, we can think of it as an **enormous collection of cooperating computers** spread geographically across the world.

Data center A $\leftrightarrow$ Data center B $\leftrightarrow$ Data center C $\leftrightarrow$...

And inside each data center, there are many cooperating machines:

server₁, server₂, server₃, ...

All connected through networking infrastructure.

Google historically operated hundreds of thousands of servers and gives a more recent estimate of **more than 2 million servers across 36 data centers**. The important point is not the exact count; it is the architectural idea. From outside, we might type `google.com` and perceive a single service. But internally, that service may involve an enormous distributed infrastructure. So we get an important contrast:

- **User’s view**: Google, a single website.
- **Physical reality**: millions of machines, networks, and data centers cooperating to provide a single service.

This is a recurring objective in distributed systems: **expose a coherent service while hiding much of the underlying distribution**. We will later give this idea a precise name: **transparency**.

 There is also another important implication. If the system contains millions of machines, failures cannot be treated as exceptional events. Suppose each individual machine is highly reliable. With millions of components, some machine

somewhere will still fail regularly. So at very large scale, **failure** is not something that might happen once in a while. It **becomes something the system must be designed to live with**. This point will become central when we discuss **partial failures** and **fault tolerance** later. For now, the main takeaway is that **a huge distributed infrastructure can behave externally like one computer/service**.

☰ Distributed systems beyond the Internet

At this point it would be easy to conclude that distributed systems are only internet applications. That would be **wrong**. The Internet, the Web, and LLM platforms are distributed systems, but many distributed systems have little or nothing to do with the public Internet.

Example 2: Distributed Systems Beyond the Internet

- **Intranets**. An organization can have a private distributed system. For example, banking systems, corporate systems, scientific computing infrastructure, server farms, grids, clouds.
For example, imagine a bank with branches in Milan, Rome, and Turin. The system is distributed even if its internal communication is largely private and not exposed to the public Internet.
- **Home networks**. Even our home can contain a small distributed environment: laptop, router, tv, NAS, phone, and so on. Different devices communicate and share resources. This is a distributed system on a dramatically smaller scale than Google, but the idea is the same: independent devices and communication.
- **Personal Area Network (PAN)**. A person's devices can themselves form a small network: phone, smartwatch, headphones, gps. For example healthcare-monitoring devices can communicate with a phone and send data to a cloud service. Again, **distribution does not imply thousands of servers. Two or three cooperating devices are already enough to introduce distributed-system concepts**.
- **Wireless Sensor Networks**. Another important category consists of many small sensing devices communicating with one another or with gateways. For example, a smart home may have many sensors and actuators communicating with a central hub. Or a smart city may have thousands of sensors monitoring traffic, pollution, or infrastructure health. These are distributed systems even if they are not part of the public Internet.

Thus a distributed system may exist across continents, inside one company, inside one building, inside one house, or even around one person. What matters is not the geographical size, but that **multiple independent computational components communicate and cooperate**.

🏠 Home and personal networks

Until now, we have mostly thought about distribution at a very large scale: Google, cloud infrastructures, LLM data centers. But a distributed system can also exist on a much smaller scale.

For example, inside a home we might conceptually have a PC, a laptop, a media device, connected through a local network and possibly communicating with remote machines through the Internet. So **distribution is not defined by physical distance**. Two computers sitting in the same room can form part of a distributed system just as computers located on different continents can.

In the previous page, we mentioned home networks and Personal Area Networks (PANs). A home network concerns devices belonging to the environment:

$$\text{home} \rightarrow \{\text{computers, entertainment devices, shared resources}\}$$

A personal-area network concerns devices surrounding a person:

$$\text{person} \rightarrow \{\text{phone, smartwatch, headphones, gps, etc.}\}$$

⚠️ The scale is small, but the distributed-system issue remains: **independent components must communicate and coordinate**.

📡 Pervasiveness of Distributed Systems

Instead of having a few clearly recognizable computers (desktop, laptop, server), **computing and communication capabilities become embedded in ordinary physical objects**. This is the basic idea behind the **Internet of Things (IoT)**. We can use three quotations to develop this idea:

1. **Computing that “disappears”**. Mark Weiser (1991) says that the most profound **technologies** eventually become **integrated into everyday life** to the point that we **stop consciously noticing them**. This is the idea of **Ubiquitous Computing** (or **Pervasive Computing**).
Think about the conceptual transition from “*use a computer*” toward “*computing is embedded in the environment around us*”. The computer itself is no longer necessarily the object we interact with.
2. **Connectivity for things**. The ITU (International Telecommunication Union) quotation captures another transition. Traditional networking gave us connectivity from anyone to anywhere (i.e., from one person to another). IoT extends this toward **connectivity for anything**. Objects that previously had no communication capabilities can contain small transceivers and participate in a network. So instead of only connecting computers, we may have a network of connected objects. The number of communicating entities consequently becomes enormous, **21.1 billion active IoT connections by the end of 2025**.
3. **From connected objects to intelligent objects**. The Negroponte quotation adds another important point. IoT is not merely attaching an

identifier to an object, such as an RFID tag that tells us where the object is. IoT is about **embedding intelligence into objects**. So a device can potentially:

- (a) Observe something about its environment (e.g., temperature, humidity, motion, light, etc.)
- (b) Process information (e.g., detect patterns, make decisions, etc.)
- (c) Communicate with other devices (e.g., send data, receive commands, etc.)
- (d) Participate in a larger system (e.g., a smart home, a smart city, etc.)

This gives us a useful progression: from ordinary objects, to connected objects, to **networked computational components**. And once we have potentially billions of these components, distributed-system techniques become essential.

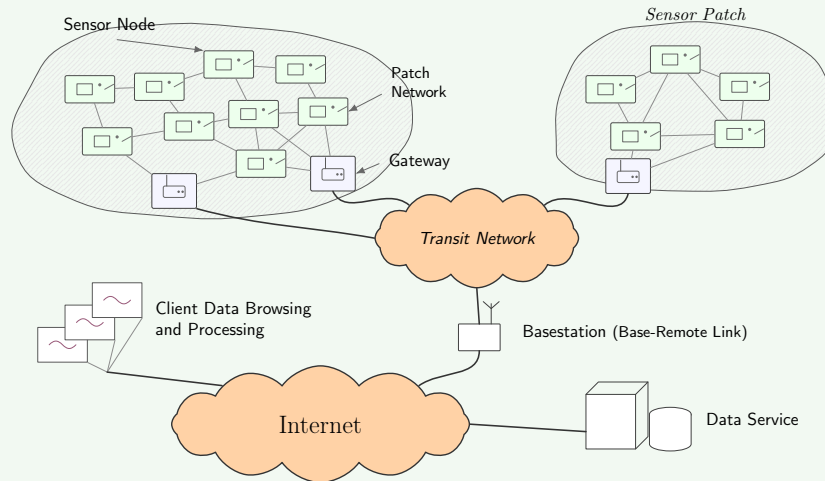
Example 3: Why a Wireless Sensor Network is a pervasive system

Imagine we want to monitor a foster, a factory, a building, or a crop field. Instead of putting one big computer in the middle, we scatter many tiny devices around the environment:

$$S_1, S_2, S_3, \dots, S_n$$

Each one may measure something like: temperature, humidity, light, vibration, pressure, motion, etc. These devices are the **sensor nodes** and they are placed on the edge of the environment.

The crucial idea is that computing becomes part of the environment itself. We are no longer just using a computer to monitor the environment. Instead, the sensors are just “there”, continuously observing the physical world. That is exactly why this is **pervasive computing**: computation is distributed throughout the environment, and it is not necessarily associated with a single device that we consciously interact with.



In the previous diagram, we represent a generic architecture for a Wireless Sensor Network (WSN).

- On the top-left, we have small devices inside the shaded region, called **sensor nodes**. Each one is a small computer with, conceptually, a CPU, memory, and a small radio transceiver. The sensor observes something physical, for example temperature, humidity, light, vibration, pressure, motion, etc.
- A group of sensor nodes may form a **Sensor Patch** (top-right), which is a small area of the environment that is monitored by a group of sensors. Inside each patch, the sensor nodes communicate through the **Patch Network**. Therefore, the sensors form a local network rather than each sensor independently connecting directly to the Internet.
- At the edge of each Sensor Patch, we have a **Gateway**. The sensor network may use some specialized local wireless technology, while the rest of the system uses a different network. The gateway bridges the two networks. **So instead of each sensor node connecting directly to the Internet, the gateway collects data from the sensor nodes and sends it to the Internet.**
- Multiple gateways connect to the **Transit Network**. This part simply transports information from the local sensor patches toward a more conventional infrastructure.
- The transit network reaches a **Base Station**. This is the point where data can be forwarded toward a larger network infrastructure. We call it also a **Base-Remote Link**, because it connects the local sensor network to a remote infrastructure.
- Once the information reaches the Internet, it can be sent to the **Data Service** shown on the bottom-right. This might conceptually be a database, a server, a cloud service, or an analytic system.

For example, suppose the sensors are measuring temperature. The data service could receive: $(t_1, 22.4^\circ C)$, $(t_2, 22.5^\circ C)$, $(t_3, 22.6^\circ C)$, and store or process those values.

- Finally, on the left of the Internet cloud, we have **Client Data Browsing and Processing**. That is simply the user/application consuming the collected data.

For example, a control software could receive the temperature data and decide to turn on a fan or an air conditioner if the temperature exceeds a certain threshold.

This example shows that distributed systems are not just servers communicating over the Internet. They can also be pervasive systems, where computation is embedded in the environment itself.

❓ Why mobility makes distributed systems harder

We can consider three different mobility scenarios that make distributed systems more challenging:

- **Normadic**. In the nomadic case, the device moves between locations, but communication is associated with whichever fixed network access point it is currently using.

For example, a laptop may move between different Wi-Fi access points in a building. Thus, the **device changes its point of attachment** to the network, and the natural question is “*how do we continue referring to the same device when its network location changes?*”. This question will become important later when we discuss *naming*.

- **Base-station mobility**. Think about a mobile phone on a cellular network. We are moving, but there is a fixed infrastructure around us. Suppose we are walking through Milan while on a call. At first our phone communicates through BS_1 , then we move far enough that another base station becomes better, so we switch to BS_2 .

The important thing is that the **device moves, but the infrastructure does not**. The network has fixed access points/base stations, and the mobile node attaches to whichever one is appropriate. The distributed-system challenge is that the device’s **point of attachment changes while the device itself remains the same entity**. So again, identity and current location must be kept separate.

- **Ad-hoc mobility**. Imagine several laptops or phones in the same area, and they can communicate directly with one another through their wireless interfaces.

$$A \leftrightarrow B \leftrightarrow C$$

If A cannot directly reach C , but B can reach both, then B may conceptually act as an intermediate node. The important feature is that **the devices themselves form the network**. There is no fixed base station organizing the connectivity.

For example, imagine several rescue teams entering an area where there is no working network infrastructure. Their devices may form a temporary network among themselves:

$$A \leftrightarrow B \leftrightarrow C \leftrightarrow D$$

Then people move. Maybe B leaves the area, so the topology changes:

$$A \leftrightarrow C \leftrightarrow D$$

Or perhaps A can no longer reach the others at all. That is why ad-hoc networks are especially dynamic: the **network topology itself changes as the nodes move**.

The three cases can be summarized as follows:

Model	Device moves?	Fixed infrastructure?
Nomadic	Yes, between attachment points.	Yes
Base-Station	Yes	Yes, explicitly through base stations.
Ad-Hoc	Yes (potentially)	No, fixed base station.

Even without mobility, a distributed system already needs to know:

- *Who is the other component?* (identity)
- *How can I reach it?* (location)

With mobility, the **second answer may continuously change**.

✓ Key Takeaways 1: Why Distributed Systems?

- Most modern applications are inherently **distributed**: their computation, data, or users span multiple communicating machines.
- Distribution is often unavoidable because of **scale**. Services such as large Web platforms and LLM infrastructures cannot realistically be provided by a single machine.
- A distributed system may involve anything from a few personal devices to millions of servers. Therefore, **distribution is not defined by geographical distance or system size**.
- Large infrastructures illustrate an important recurring idea:

Many independent components $\longrightarrow$ one coherent service

The user may perceive a single service even though the implementation is spread across many machines. This anticipates the concept of **transparency**.

- At sufficiently large scale, **failures are normal rather than exceptional**. A distributed system must therefore be designed assuming that individual components may fail while the rest of the system continues operating.
- Distributed systems are not limited to the Internet or the Web. Examples include: intranets, home networks, personal area networks (PANs), Internet of Things (IoT) systems, and wireless sensor networks (WSNs).
- **Pervasive/ubiquitous computing** pushes computation into ordinary objects and environments. IoT extends connectivity from people and computers toward potentially huge numbers of **networked intelligent objects**.
- A **Wireless Sensor Network** is a representative pervasive distributed system: sensor nodes form a local patch network, which connects to a gateway, which connects to a transit network, which connects to the Internet or a data service.

The gateway bridges the local sensor network with the larger communication infrastructure.

- **Mobility makes distribution harder** because the identity of an entity and its current network location must be separated:
 - **Nomadic**: the device moves between different attachment points.
 - **Base-station**: the mobile device communicates through a fixed infrastructure and may switch from one base station to another.
 - **Ad-hoc**: there is no fixed infrastructure; the mobile nodes themselves form the network, so the network topology may change as they move.

1.1.2 Why Distribute a System?

Economics

Historically, computing was often organized around large centralized machines, such as **mainframes** (i.e., a single computer that serves multiple users), with many users or terminals depending on that central machine. The alternative is to use many smaller machines $M_1, M_2, M_3, \dots$ and let them cooperate. The economic argument is based on the **price/performance ratio**.

Instead of buying one extremely powerful machine, it **may be more convenient to buy several cheaper machines and combine their resources**. The idea is not that several small computers automatically behave like one large computer. We need software that coordinates them. But **distribution allows us to exploit the aggregate resources of many machines**. So the economic motivation is that **it can be cheaper to obtain computing power by combining multiple machines**. This is one of the historical reasons why distributed computing became attractive.

Incremental growth and load balancing

This is one of the most important practical advantages. Suppose a centralized system starts with one machine M_1 and its workload increases. Eventually:

$$\text{load} > \text{capacity of } M_1$$

With a distributed architecture, we can add more machines:

$$M_1 \rightarrow M_1 + M_2 \rightarrow M_1 + M_2 + M_3 \rightarrow \dots$$

This is **incremental growth**. We do not necessarily need to replace the entire system with a much larger machine every time demand increases. Instead, **capacity can grow by adding resources**. This is closely related to **load balancing**.

Imagine that three machines are available:

$$M_1, M_2, M_3$$

And requests arrive:

$$R_1, R_2, R_3, R_4, R_5, R_6$$

Rather than sending everything to M_1 :

$$R_1, R_2, R_3, R_4, R_5, R_6 \rightarrow M_1$$

We can distribute them:

$$R_1, R_2 \rightarrow M_1, \quad R_3, R_4 \rightarrow M_2, \quad R_5, R_6 \rightarrow M_3$$

The objective is to use the available resources more uniformly.

So there are really two connected ideas:

- **Incremental growth:** add resources gradually as the system grows.
- **Load balancing:** spread the workload among the available resources.

In summary, it makes easier to **evolve the system** and **use its resources uniformly**.

Inherent distribution

Sometimes we distribute a system because we want the benefits of distribution. But in other cases, **distribution is not really a design choice, but rather the problem itself is already distributed.**

For example, consider a bank. Its users, branches, ATMs, databases, and services may naturally exist in different locations: ATM Milan, ATM Rome, ATM New York, etc. All of them interact with the same overall banking system. Trying to force everything into one physical place would not make sense (i.e., it would be inefficient and impractical). This is what **inherent distribution** means: **the entities involved in the problem are naturally located in different places.**

Reliability

Distribution can also improve **reliability**. Consider a purely centralized architecture client-server. If that server fails, the entire service may disappear. This is a **single point of failure**. Now suppose the system is distributed over several components $S_1, S_2, S_3, \dots$. If one of them fails, the others may still be able to provide the service. So **one failure does not necessarily bring down the whole system.** This is particularly important for **mission-critical applications**, where complete service failure is unacceptable.

But there is an important subtlety. Distribution does **not automatically guarantee reliability**. It merely gives us the possibility of designing a system that survives individual failures. For example, if all machines depend on one central database, then the database may still be a single point of failure. Thus, to obtain reliability, we often need techniques such as **redundancy, replication, failure detection, and recovery**. These are all topics that will be covered in later sections. However, here the main point is that **distribution allows failures to be partial rather than necessarily total.** That sounds good, and it often is, but partial failures will also turn out to be one of the hardest aspects of distributed systems.

The complexity cost of distribution

Obviously, distribution is not free. **Distribution systems are considerably more difficult to get right.** Everything we have discussed sounds attractive: lower cost, easy growth, load balancing, reliability, and so on. But none of this is free.

Consider two components A and B on two different machines. If A needs something from B , communication now happens through a network:

$$A \xrightarrow{\text{network}} B$$

That introduces many questions:

- *What if the message is delayed?*

$$A \xrightarrow{\text{delayed}} B$$

- *What if the message is lost?*

$$A \not\xrightarrow{\text{lost}} B$$

- *What if B has crashed?*
- *What if B is alive but just very slow?*

From *A*'s perspective, slow network and failed machine may be difficult to distinguish. Then there are **synchronization problems**.

Suppose two machines modify the same logical data:

$$A : x \leftarrow 5$$

$$B : x \leftarrow 7$$

Which value should the system consider correct? There is also no single perfectly shared notion of time:

$$\text{clock}_A \neq \text{clock}_B$$

And components may fail independently.

So distributing a system trades one kind of difficulty for another. A centralized system may be easier to reason about one machine at a time, while a distributed system gives us additional capabilities (many machines) but also introduces new problems: communication, concurrency, partial failure, coordination, and so on. This is the central trade-off.

✓ Key Takeaways 2: Why Distribute a System?

There are four main motivations for distributing a system:

1. **Economics**. Several smaller machines may provide a better **price/performance ratio** than one extremely powerful centralized machine.

many cheaper machines $\longrightarrow$ aggregate computational resources

2. **Incremental growth and load balancing**. A distributed system can grow gradually by adding resources as the workload increases:

$$M_1 \rightarrow M_1 + M_2 \rightarrow M_1 + M_2 + M_3 \rightarrow \dots$$

The workload can then be distributed among those resources rather than concentrating it on a single machine.

3. **Inherent distribution.** Sometimes the problem itself is naturally distributed: users, resources, or activities already exist at different locations.

distributed problem $\implies$ naturally distributed solution

Typical examples are banking systems, reservation systems, distributed games, collaborative applications, and mobile applications.

4. **Reliability.** Distribution makes it possible for the system to continue operating even if one component fails:

failure of one component $\nRightarrow$ failure of the entire system.

This is especially important for **mission-critical systems**. However, reliability is not automatic: the architecture must avoid introducing new single points of failure.

So in general, **distribution brings advantages, but it also makes systems harder to build correctly.** Communication, coordination, concurrency, and failures introduce complexity that does not appear in the same form in centralized systems.

References

- [1] Margara Alessandro Cugola Gianpaolo Saverio. Distributed systems. Slides from the HPC-E master's degree course on Politecnico di Milano, 2026.

Index

P

Pervasive Computing 8

U

Ubiquitous Computing 8